

Enteroviruses

**Background document for the
WHO Guidelines for drinking-water quality and the
WHO Guidelines on sanitation and health**

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Contents

Highlights.....	1
1 General description.....	2
2 Human health effects	2
3 Epidemiology	3
4 Transmission	3
5 Detection methods.....	4
6 Source and occurrence	5
6.1 Faecal pollution sources.....	5
6.2 Surface water	6
6.3 Groundwater	6
6.4 Drinking-water.....	6
6.5 Seawater.....	7
6.6 Soil and crops.....	7
7 Significance of sanitation.....	7
8 Significance in drinking-water	7
9 Prevention and management	8
9.1 Sanitation systems.....	8
9.2 Drinking-water supplies.....	8
10 ICD-11 classification.....	9
Selected references.....	10
Annex. Approach to content development and declarations of interest	14

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Highlights

- There is strong evidence linking transmission of enteroviruses with inadequate sanitation, and poor hygiene. Enteroviruses are significant waterborne pathogens, but the evidence linking transmission through drinking-water is moderate. Outbreaks are primarily associated with recreational waters.
- Humans are the only known reservoir for disease-causing strains of enteroviruses in humans.
- Transmission is primarily by the faecal–oral route and through inhalation of airborne viruses. Fomites, food and soil contaminated by infected faeces are important routes of transmission. Oral–oral spread may also occur. Waterborne transmission of enteroviruses has been confirmed for only two outbreaks, and these were associated with recreational water.
- The spectrum of diseases caused by enteroviruses is broad and ranges from mild febrile to severe neurological illness. Enteroviruses can cause highly persistent infections. The prevalence of asymptomatic infection is extremely high, and reinfection with the same virus often results in milder symptoms, subclinical or asymptomatic infection.
- Enterovirus infections are common but under-reported globally. Children under 5 years of age are the most at-risk group for infections. In temperate regions, enteroviral infection is a seasonal illness, with the peak of cases occurring in the summer while occurring all year round in tropical areas.
- The relative infectivity of most enteroviruses has not been determined, but similar to other enteric viruses, enteroviruses are probably highly infective.
- Standard cell culture, as well as molecular-based methods, are available for the detection of enteroviruses in environmental samples.
- Enteroviruses are commonly present in raw sewage, treated wastewater, drinking-water sources and occasionally in inadequately treated drinking-water. Persistence in water is likely to vary among species but in general, can be considered long.
- Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains are needed to manage microbial risks, including from enteroviruses. However, as they spread easily person-to-person contact, airborne transmission and fomites, other control measures, including adequate hygiene, are required.
- Reduction of enteroviruses by wastewater and drinking-water treatment varies depending on the process. Enteroviruses have low to moderate resistance to disinfection with chlorine, but chlorine can be effective provided adequate doses and contact times are applied.
- Due to the smaller size of enteroviruses relative to *E. coli*, resulting in lower removal by treatment processes such as filtration, and their greater environmental persistence, the absence of *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator of the absence of human enteroviruses in drinking-water. However, the presence of *E. coli* is a suitable indicator of the possible presence of human enteroviruses.

1 General description

The genus *Enterovirus* is a member of the Picornaviridae family. According to a recent reorganization of virus taxonomy, the genus contains enterovirus species A through L and rhinovirus species A, B and C (ICTV, 2021). Over 100 serotypes that infect humans are identified within these 12 enterovirus species, among which are poliovirus, coxsackievirus and echovirus, which retain their old taxonomic designations as common names. When discussing enteroviruses, a distinction is often made between poliovirus (*Enterovirus* C) and non-polio enteroviruses.

Enteroviruses are among the smallest known viruses and consist of a single-stranded RNA genome in a non-enveloped icosahedral capsid with a diameter of 20 nm to 30 nm (Zaoutis & Klein; 1998; Fong & Lipp, 2005; Palacios & Oberste, 2005).

Most enterovirus strains are human-specific, but other species of the genus infect animals other than humans. For instance, simian enteroviruses are thought to have been derived originally from human-specific coxsackieviruses. However, none of these non-human-specific forms have been known to infect humans (Palacios & Oberste, 2005).

2 Human health effects

Enterovirus infection originates in the gastrointestinal tract and can cause a broad spectrum of diseases. Mild illness characterized by cough, rash and fever is the most generic symptom of enterovirus infection. However, enteroviruses are also associated with severe neurological illness, including aseptic meningitis, encephalitis and poliomyelitis and respiratory illnesses (Sinclair & Omar, 2022). Aseptic meningitis appears to be the most common severe clinical manifestation associated with enterovirus infections in children and adults, especially in high-income countries (Fong & Lipp, 2005; Pallansch et al., 2013; Percival et al., 2013; Tapparel et al., 2013). Polioviruses may infect the central nervous system. Initial symptoms of infection are fever, fatigue, headache, vomiting, stiffness of the neck and pain in the limbs. One in 200 infections causes irreversible acute flaccid paralysis. The maximum extent of paralysis is reached between a few days to 2 weeks. Limbs, particularly legs, are most often affected, but paralysis of the respiratory muscles can occur and may result in death (WHO, 2022a). Chronic conditions such as muscular atrophy and often deformity, dilated cardiomyopathy and chronic fatigue syndrome follow paralysis. Non-polio enteroviruses can cause “poliomyelitis-like” infections, including acute flaccid paralysis (Betancourt & Shulman, 2017). Coxsackievirus type B may target the respiratory system and even contribute to diabetes (Yang et al., 2021) or cardiovascular diseases (Zhang et al., 2023). Echoviruses are generally associated with respiratory complications (Fong & Lipp, 2005). Enterovirus 71 may cause neurologic symptoms and has been associated with hand-foot-and-mouth disease (Palacios & Oberste, 2005; Pallansch et al., 2013; Chang et al., 2016). Mortality rates from clinical infection can be as high as 10% (Betancourt & Shulman, 2017).

The prevalence of asymptomatic infections can be extremely high for enteroviruses, as high as 99% for poliovirus (Zaoutis & Klein, 1998). Further, re-infection with the same virus often results in milder symptoms or subclinical or asymptomatic infection (Zaoutis & Klein, 1998).

3 Epidemiology

Enteroviruses are a common but under-reported cause of human infections globally. With the exception of poliomyelitis, the disease burden is poorly estimated (Betancourt & Shulman, 2017). Children under 5 years of age constitute the most at-risk group for infections (Zaoutis & Klein, 1998; Pallansch et al., 2013). Enteroviruses exhibit both epidemic and endemic patterns of incidence (Palacios & Oberste, 2005). Seasonally, infections from enteroviruses have tended to occur in summer in countries in the temperate areas of the northern hemisphere (e.g. United States of America (USA)). In contrast, in tropical areas, infections occur all year round (Zaoutis & Klein, 1998).

One review (Brouwer et al., 2021) found that *Enterovirus B* was the most detected species worldwide. *Enterovirus C* was more often detected in Africa, and *Enterovirus A* was more often detected in Asia. Echovirus 30 was the most common type detected. The global prevalence of poliovirus has decreased due to worldwide eradication efforts through vaccination, but wild poliovirus type 1 continues to be detected in parts of Afghanistan, Pakistan and Mozambique (WHO, 2023a). Novel enteroviruses (e.g. D68, enterovirus 71) continue to emerge and may become new public health threats (Qiu, 2008; Holm-Hansen et al., 2016; Fall et al., 2022).

Transmission

4 Transmission

Enteroviruses are transmitted by the faecal–oral route and by inhalation of airborne viruses (Muehlenbachs et al., 2015). The relative importance of the different modes of transmission probably varies with the particular enterovirus and environmental settings. The faecal–oral pathway through person-to-person contact predominates in areas with poor sanitary conditions, while transmission through inhalation predominates in higher income settings. Oral–oral spread by contact with contaminated saliva may be an exposure pathway. Transmission through contaminated fomites and soil are considered important (Palacios & Oberste, 2005; Betancourt & Shulman, 2017).

Waterborne transmission of enteroviruses has been epidemiologically confirmed for only two outbreaks, and these were associated with recreational water settings. One was associated with children bathing in a lake in the USA (Hawley et al., 1973), while another outbreak occurred among travellers to Mexico after swimming in sewage-contaminated seawater (Begier et al., 2008). Drinking-water represents a likely source of enterovirus infection. The limited knowledge on the role of waterborne transmission could be related to several factors, including the dominance of person-to-person spread, wide range of clinical symptoms, frequent asymptomatic infection and the diversity of serotypes.

The infective doses have not been determined for most species of enteroviruses, and reported studies have yielded varying results. Duncan & Edberg (1995) reported that the tissue culture infective doses for human enteroviruses may be as low as one to four infective particles. A challenge study using human volunteers estimated the ID₅₀ (50% infective dose, i.e. the number of organisms required to infect 50% of exposed subjects) for echovirus 12 to be 1.1×10^3 virus particles (Schiff et al., 1984; Teunis et al., 1996). In contrast, Betancourt & Shulman (2017)

noted that echoviruses are highly infective. Overall, based on these results and the high relative infectivity of other enteric viruses, enteroviruses are also considered to be highly infective.¹

Most enterovirus have incubation periods of 3 to 10 days on average (Zaoutis & Klein, 1998), although incubation periods for polio vary from 5 to 30 days (with an average of 10 days). Poliovirus can be transmitted for 2 days to 6 weeks after exposure (Webber, 2005). Viruses tend to be shed in stools for 10 days to 2 months after exposure and may be present in mucous in the throat from 10 days to 22 days post-exposure (Zaoutis & Klein, 1998). Immunodeficient infected persons may shed for longer: from 10 months to 28 years for some individuals (Chung et al., 2001). Enteroviruses are excreted in the faeces of infected individuals in high concentrations; up to 10^6 to 10^7 viral particles/g of faeces have been reported (Betancourt & Shulman, 2017). Poliovirus vaccines have led to the eradication of polio in most countries and provide long-term protection. Residual antibodies are detected for months to years after exposure (Zaoutis & Klein, 1998). Residual immunity tends to be present post-infection, but if reinfection in a previously infected individual occurs, symptoms are reduced, and the illness may be subclinical (Zaoutis & Klein, 1998).

5 Detection methods

A number of enteroviruses (poliovirus, coxsackievirus B, echovirus and enterovirus) are readily detected by cytopathogenic effect in cell cultures (Haramoto et al., 2018; Corpuz et al., 2020). However, the use of multiple cell lines is essential to improve the detection of the different types of enteroviruses. Molecular methods have been developed to detect serotypes that do not readily grow in cell culture and for unculturable serotypes (e.g. enterovirus 5). However, distinguishing between viable and non-viable viruses is a shortcoming of molecular methods. Combining cell culture infectivity and molecular methods make it possible to improve the detection of infectious enteroviruses. US EPA Method 1615 is a standard method for quantifying enteroviruses in water, using both culture and molecular approaches (Fout et al., 2014). Integrated methods have been used in combination with viral metagenomics approaches for cell culture evaluation of the virological quality of wastewater (Aw et al., 2014).

¹ From experiments with human volunteers, from epidemiological evidence and from experimental animal studies. Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: High means infective doses can be $1-10^2$ organisms or particles, Moderate 10^2-10^4 and Low $> 10^4$.

6 Source and occurrence

Various environmental matrices have been examined for enteroviruses. A summary of their occurrence, concentrations and persistence is provided in Table 1 and the subsequent text.

Table 1. Summary of enterovirus occurrence data (as reported in Betancourt & Shulman (2017), unless otherwise noted)

Source	Prevalence		Concentration ^a (GC/L or PFU/L)				Method ^b
	Studies (samples)	% positive	Studies (samples)	Average	Studies (samples)	Range	
Raw sewage	27 (8169)	86	3 (164)	1.1×10 ⁵	12 (2788)	ND ^c –2.1×10 ⁷	qPCR
	1(74) ^d	84	1(74)	968	1(74)	ND–3.7×10 ⁴	culture
Treated wastewater	4 (150)	69	3 (79)	3.4×10 ⁴	3 (79)	3.5×10 ¹ –2.5×10 ⁵	qPCR
	1(89) ^d	30	1(89)	0.017	1(89)	ND–0.39	culture
Wastewater sludge	3 (34)	100	3 (34)	6.6×10 ⁶	2 (22)	<4×10 ⁴ – 3.8×10 ⁷	qPCR
Surface water	1 (16)	63	NR	NR	1 (16)	ND–9.8×10 ³	qPCR
	1(58) ^d	75	1(58)	0.41	1(58)	ND–10	culture
	1(75) ^e	75	NR	NR	1(75)	3.3×10 ^{–3} –5.2	culture
Groundwater	1 (1204)	9	1 (1204)	0.8	1 (1204)	ND–850	qPCR
Drinking-water			1(8)	1.6–33			culture
	1(12) ^f	8.3	1(12)	1.7	1(12)	ND–13	qPCR

^agenome copies or plaque forming units/L

^bquantitative polymerase chain reaction

^cnot detected

^dCostán-Longares et al., 2008

^eLodder et al., 2010

^fVarughese et al., 2018

6.1 Faecal pollution sources

Enteroviruses are commonly present in raw sewage worldwide, often in 100% of samples and in concentrations up to 2.1×10^7 genome copies (GC)/L (Betancourt & Shulman, 2017; Haramoto et al., 2018). In one study (Hewitt et al., 2011) detection rates and concentrations in raw sewage varied depending on the size of wastewater treatment plant and ranged from 33% and 10³ GC/L for small plants (serving less than 4000 people) to 89% and 10⁷ GC/L for large plants (serving greater than 130 000 people). Enteroviruses were present in treated wastewater with detection rates and concentrations ranging from 11% and 10² GC/L for small plants to 78% and 10⁵ GC/L for large plants. A study in Spain showed that 84% of wastewater samples contained culturable enteroviruses with a geometric mean of 968 plaque forming units (PFU)/L (Costán-Longares et al., 2008). Fifty-one percent of the secondary treated and 30% of the tertiary treated wastewater samples contained enteroviruses with geometric means of 0.36 and 0.017 PFU/L, respectively. Shi et al. (2021) did not detect any enteroviruses in samples of treated wastewater effluent from a city in China,

In several studies, the occurrence of enteroviruses and their concentrations were determined in various types of sewage sludge (Corpuz et al., 2020; Gholipour et al., 2022). Wong et al. (2010) detected enteroviruses in 42% of mesophilic anaerobic digested samples (n=12) and 100% of nondigested dewatered sludge samples (n=3). Bibby & Peccia (2013) detected enteroviruses in 70% of mesophilic anaerobic digested samples (n=10). Jebri et al. (2012) detected enteroviruses in 7.7% of activated sludge samples. Estimated concentrations of 10² to 10⁴ GC/g (dry weight)

were reported for primary, activated and anaerobically digested sludges (Betancourt & Shulman, 2017).

In laboratory experiments, polioviruses showed moderate persistence in unfiltered primary effluent with 3 log₁₀ inactivation recorded after 11 days at 23 °C; persistence was short in secondary effluent with 3 log₁₀ inactivation recorded after 5.7 days at 23 °C (Gundy et al., 2009). Some evidence is emerging that enteroviruses (along with other human viruses) remain infectious within free-living amoebae and persist within their cysts and extracellular vesicles (Atanasova et al., 2018).

6.2 Surface water

Enteroviruses have been detected frequently in surface water (Chapron et al., 2000; Varughese et al., 2018) and flood waters (de Man et al., 2014), especially those affected by human sewage (Rodríguez-Díaz et al., 2009). A study in the Netherlands detected enteroviruses (0.0033–5.2 PFU/L) in 75% of the 75 surface water samples used for drinking-water production (Lodder et al., 2010). A study in Spain showed that 75% of the river samples contained culturable enteroviruses with a geometric mean of 0.41 PFU/L (Costán-Longares et al., 2008). Shi et al. (2021) detected enteroviruses in 91.7% of samples from two urban rivers in China. In fresh water sources, natural inactivation rates depend on temperature, with 1 log₁₀ reductions taking between 1 and 1.5 days at temperatures between 12 °C and 20 °C, while reductions of 6.5 to 7 log₁₀ may take 8 weeks at 22 °C (Betancourt & Shulman, 2017). In a study by de Roda Husman et al. (2009), infectious poliovirus 2 was not detected in artificial surface water stored at 22 °C after 240 days, compared with 150 and 120 days for poliovirus 1 and coxsackievirus B4, respectively.

6.3 Groundwater

Data on the occurrence of enteroviruses in groundwater are limited. In one study of 14 communities using non-disinfected groundwater in the USA, enteroviruses were detected in 9% of samples (n=1204) (Borchardt et al., 2012). The mean and maximum concentrations were 0.8 GC/L and 850 GC/L, respectively. Poliovirus has been shown to remain infectious for 90 days at 10 °C in groundwater (Betancourt & Shulman, 2017). In another study (de Roda Husman et al., 2009), coxsackievirus B4 was no longer detected in 22 °C artificial ground water after 110 days, whereas polioviruses 1 and 2 were not detectable after 160 and 150 days, respectively.

6.4 Drinking-water

Enteroviruses have been detected in drinking-water samples in several studies. For example, one study detected human enteroviruses in 43% of tap water samples in Karachi, Pakistan (Rashid et al., 2021). Although usually present in low concentrations (Varughese et al., 2018) one study reported up to 33 PFU/L (Betancourt & Shulman, 2017). Enteroviruses have also been found within ubiquitous tap water amoebae. For example, infectious coxsackieviruses have been found in all life stages of the amoebae *Vermamoeba vermiformis* (Atanasova et al., 2018). Enterovirus have been shown to remain infectious for over a year in mineral water at low temperatures (4 °C), yet less than a month at higher temperatures (25 °C) (Fong & Lipp, 2005). In laboratory experiments poliovirus survived for long periods in tap water with 3

$\log_{10}$ inactivation achieved after 71 days at 23 °C (Gundy et al., 2009). Based on survival in a variety of waters the persistence of enteroviruses can be considered long.²

6.5 Seawater

Enteroviruses have been detected in samples of coastal waters – often in a high proportion (up to 100%) of samples depending on the setting (Fleisher et al., 1996; Betancourt & Shulman, 2017). Concentrations vary but can reach 10^3 or 10^4 GC/L, and in one study concentrations up to 6.3×10^7 GC/L were reported (Zhang et al., 2010). A study in Spain (Costán-Longares et al., 2008) showed that in 63.3% of seawater samples contained culturable enteroviruses with a geometric mean of 0.36 PFU/L. Several studies have demonstrated enterovirus occurrence in shellfish (Betancourt & Shulman, 2017). The time taken for a 2 $\log_{10}$ natural reduction in poliovirus 1 ranges from 6 days in 5 °C seawater to 3 days in 25 °C seawater (Betancourt & Shulman, 2017).

6.6 Soil and crops

Both soil and crops can be contaminated with enteric viruses through irrigation with contaminated water or the application of organic waste and biosolids. Enterovirus persistence in soil and on crops is relatively short: less than 20 days in some circumstances, although it may be up to 12 weeks or more in cool conditions (Santamaria & Toranzos, 2003; Rzezutka & Cook, 2004). Enterovirus have also been detected in river sediments (Ali et al., 2004; Garcia-Aljaro et al., 2017). Tests of specific fruits and vegetables have revealed a survival time of about 18 to 36 days (Rzezutka & Cook, 2004).

7 Significance of sanitation

Safe sanitation systems are important barriers against the spread of enteroviruses (Betancourt & Shulman, 2017). However, given the importance of person-to-person and airborne transmission, additional control measures including good personal hygiene, such as hand washing, are needed to effectively prevent transmission.

8 Significance in drinking-water

There is moderate evidence linking transmission of enteroviruses with unsafe drinking-water. An epidemiological investigation of a 1982 poliomyelitis outbreak (Kim-Farley et al., 1984) indicated that children were five times more likely to become ill if they received water from non-municipal rather than municipal sources. A multivariate analysis of a 2010 poliomyelitis outbreak in the Republic of Congo (Gregory et al., 2012) identified the use of a well for drinking-water during a water shortage as one of the risk factors for illness. A case-control study of an enterovirus outbreak in Belarus in 2003 indicated that illness was associated with drinking-water from a community water system (Amvrosieva et al., 2006). The laboratory

² Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: Detection period for infective stage in water at 20 °C: Short, up to 1 week; Moderate, 1 week to 1 month; Long, over 1 month.

examination indicated a high level of genetic similarity between waterborne and clinical isolates.

9 Prevention and management

Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains are needed to manage microbial risks, including from enteroviruses.

Treatment processes based on size exclusion (e.g. filtration through media or soil in septic tank drainage areas) or settling in primary wastewater treatment processes will be limited for removing viruses, including enteroviruses. Enteroviruses have low to moderate resistance to oxidizing chemical disinfectants such as chlorine but have low resistance to disinfection by UV light.

9.1 Sanitation systems

Sanitation safety plans (WHO, 2022b) should include control measures to safely collect, treat and discharge or reuse wastewater and faecal sludges. Removal or inactivation of enteroviruses by wastewater treatment processes is variable, and based on available evidence, the reduction of viruses by treatment processes is likely to be lower than the reduction of faecal indicator bacteria such as *E. coli*.

Data on the reduction of specific members in the *Enterovirus* genus in on-site systems are limited and have been summarized by Betancourt & Shulman (2017). Reported reductions were 3 log₁₀ in waste stabilization ponds and 2 to 3 log₁₀ in constructed wetlands for enteroviruses, and 3 log₁₀ reduction of polioviruses in oxidation ponds. For centralized sewage treatment, viruses are less effectively removed by size exclusion or settling processes, compared to larger pathogens. Compared to other viruses, there is a reasonable amount of data on removal of enteroviruses by sewage treatment (Betancourt & Shulman, 2017). The removal of enteroviruses in primary treatment may reach 1 log₁₀ post-flocculation. Reported removals of enteroviruses in sewage treatment systems range from 0.19 to 3 log₁₀ for activated sludge plants and less than 1 to 4.7 log₁₀ for membrane bioreactors (Betancourt & Shulman, 2017). However, enteroviruses have been detected in treated wastewater at concentrations of 3.5×10^1 to 2.5×10^5 GC/L and some enteroviruses removed from wastewater during treatment likely remain viable in the sludge.

9.2 Drinking-water supplies

Within a water safety plan (WHO 2012; 2023b), control measures to manage potential risks from enteroviruses in drinking-water should include protection of source waters (surface or groundwater) from contamination by human faecal wastes, application of adequate treatment processes and protection of drinking-water quality during distribution and storage.

It is likely that the removal of enteroviruses by conventional drinking-water treatment processes (coagulation, flocculation, sedimentation and filtration) will be similar to that reported for other enteric viruses (Teunis et al., 2009). Moreover, conventional treatment can remove 1.5 log₁₀ of noroviruses (Shin & Sobsey, 2015), which are also non-enveloped RNA viruses of similar size

as enteroviruses. Enteroviruses vary in sensitivity to free chlorine, with some having low resistance while others, including coxsackievirus B5, having moderate resistance.³ Moreover, resistance may be influenced by the different receptors on various host cells used to study infectivity; echovirus 11 inactivation was impacted two-fold depending on the host cell (Torii et al., 2023). However, chlorine can still be effective, provided adequate doses and contact times are applied. Reported Ct (product of disinfectant concentration and contact time) values for 2 log₁₀ inactivation by free chlorine at 5 °C and pH 8 are 0.5 mg.min/L for echovirus 11, 0.99 mg.min/L for echovirus 1, 0.65 mg.min/L for coxsackievirus B3 and 4.7 mg.min/L for coxsackievirus B5, (Cromeans et al., 2010). The Ct value for 2 log₁₀ inactivation of poliovirus at pH 7.5 at 5 °C was reported at 1.4 mg.min/L (Black et al., 2009). Enteroviruses have low resistance to UV light. The reported doses necessary for a 2 log₁₀ inactivation are 17 mJ/cm² for echovirus 1, 14 mJ/cm² for echovirus 11, 16 mJ/cm² for poliovirus 1, 16 mJ/cm² for coxsackievirus B3, and 18 mJ/cm² for coxsackievirus B5 (Gerba et al., 2002). However, the association of enteroviruses (as well as probably other viruses) with cells compared to being dispersed can significantly increase UV resistance (Park et al., 2011).

Due to the smaller size of enteroviruses relative to *E. coli*, resulting in lower removal by treatment processes such as filtration, and their greater environmental persistence, the absence of *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator of the absence of human enteroviruses in drinking-water. However, the presence of *E. coli* is a suitable indicator of the possible presence of human enteroviruses.

10 ICD-11 classification

1C8E.1	Enteroviral meningitis
1C8Y	Encephalitis due to Enterovirus
1D02.1	Myelitis due to Enterovirus
1D91	Enterovirus infection of unspecified site
1F05.2	Enteroviral exanthematous fever

³ Within pathogen species and groups, there are likely to be variations in resistance, which could be further impacted by characteristics of the water supply and operating conditions. Resistance is based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: 99% inactivation at 20 °C where, generally, Low represents a Ct₉₉ of < 1 min.mg/L, Moderate 1–30 min.mg/L and High > 30 min.mg/L (where C = the concentration of free chlorine in mg/L and t = contact time in minutes) under the following conditions: the infective stage is freely suspended in water treated at conventional doses and contact times, and the pH is between 7 and 8.

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Annex. Approach to content development and declarations of interest

The background documents were developed drawing from the following resources (i) microbial fact sheets from the fourth edition of the WHO Guidelines for Drinking-Water Quality (ii) pathogen-specific chapters in the Global Water Pathogen Project (GWPP) book “*Sanitation and Disease in the 21st Century*”, an update of the seminal book by Feachem et al. 1983 “*Sanitation and Disease: Health Aspects of Excreta and Wastewater Management*” and (iii) a rapid review of the literature at the start of the development of the background documents and before finalization in December 2023, focusing on identifying additional recent high quality literature reviews on a given pathogen. The documents were prepared by members of the WHO drinking-water quality microbiology group as well as external scientists with expertise in the relevant pathogens. The documents went through several iterations based on inputs from members of the WHO drinking-water quality microbiology working group and WHO technical staff. In addition, each document was reviewed by at least two external peer reviewers including authors of the GWPP book chapters. The extensive review process was designed to ensure technical accuracy and to identify any data gaps. The background documents were discussed at three expert meetings to resolve any issues and ensure the content reflected a consensus of expert opinions. Upon satisfactory resolution of any further data gaps or issues, the background documents went through technical editing before finalization.

Before the key expert meetings mentioned above, and for this work more broadly, all microbiology working group members submitted declarations of interest to WHO. These were to disclose any potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of the guidance. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects discussed at the meetings or covered by the guidance.